Scrolling or Comprehending? Exploring the Impact of Digital Media Consumption on Deep Reading Engagement Among Tertiary Learners

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Abstract

In an era where digital technology increasingly mediates educational experiences, this study investigates how digital media consumption influences deep reading engagement among tertiary learners. As students shift from print to digital texts, concerns have emerged regarding the potential decline in critical reading skills, sustained attention, and comprehension. This study used a qualitative research design to explore ten diploma-level students' reading behaviours and perceptions at Politeknik Kota Kinabalu, Sabah. Data were collected through open-ended survey questions and analysed using thematic analysis to identify student engagement patterns with academic versus digital media texts and the factors affecting their concentration in digital reading environments. Findings reveal that while digital media is perceived as more engaging and convenient, it often promotes surface-level reading and limits deep comprehension. In contrast, though more cognitively demanding, academic texts foster better retention and critical analysis when students apply active reading strategies. Key factors influencing reading engagement include environmental distractions, emotional well-being, content relevance, and self-regulation techniques such as the Pomodoro method and digital annotation tools. This study contributes to the growing body of literature on digital literacy by foregrounding student voices and highlighting the complex interplay between technology, cognition, and motivation. The findings have important implications for English language learning, emphasising the need to balance digital accessibility with pedagogical strategies that support deep reading. Educators are encouraged to create structured, student-centred environments that nurture reflective and focused engagement with texts in both digital and print formats.

Keywords**:** *deep reading, digital media, reading engagement, tertiary learners*

Introduction

The proliferation of digital media has transformed the reading habits of tertiary learners in the digital age. The convenience and accessibility of digital texts have led to an increased reliance on screens for both academic and leisure reading. However, this shift raises concerns about its impact on deep reading engagement a cognitive process involving sustained attention, critical analysis, and reflection. Deep reading is essential for academic success, yet the nature of digital media consumption may be altering how students engage with texts.

Digital media consumption has become ubiquitous among university students, with many preferring digital formats over traditional print. Integrating hyperlinks, multimedia, and interactive features in digital texts offers a dynamic reading experience. Nevertheless, these features may contribute to superficial reading practices. For instance, Delgado et al. (2018) found that students reading digital texts exhibited lower comprehension levels than those reading print, particularly in tasks requiring deep understanding. Similarly, Mangen et al. (2019) reported that digital reading often reduces recall and inferential reasoning.

The cognitive processes involved in digital reading differ from those in print reading. Digital texts often encourage skimming and scanning, which can impede the development of deep reading skills. Wolf (2018) argues that the brain's circuitry for deep reading is not naturally activated when engaging with digital texts, potentially affecting critical thinking and empathy. Moreover, the constant distractions inherent in digital environments, such as notifications and hyperlinks, can fragment attention and hinder sustained focus (Carr, 2010).

The central issue motivating this study is the potential decline in deep reading engagement among tertiary learners due to increased digital media consumption. While digital texts offer convenience, growing evidence shows they may not support the cognitive processes required for deep comprehension. This is particularly concerning in higher education, where students must engage critically with complex materials. The shift towards digital reading may inadvertently compromise students' ability to analyse, synthesise, and evaluate information effectively.

Furthermore, perceived ease of access and multitasking capabilities influence the preference for digital reading. However, these perceived benefits may come at the cost of reduced engagement and comprehension. For example, a study by Clinton (2019) found that students who read digitally tended to overestimate their understanding of the material, leading to poorer performance in comprehension assessments. This overconfidence may stem from the illusion of knowledge that digital texts can create, where the ease of access to information is mistaken for actual understanding.

Despite the growing literature on digital reading, a gap exists in understanding how digital media consumption affects deep reading engagement among tertiary learners. Most existing studies focus on general reading comprehension or compare digital and print reading without delving into the nuances of deep reading processes. Moreover, there is a lack of research examining the long-term implications of digital reading habits on students' academic performance and critical thinking skills.

Additionally, cultural and contextual factors influencing digital reading practices among tertiary learners are underexplored. For instance, the impact of educational systems, technological infrastructure, and students' digital literacy levels on reading engagement warrants further investigation. Understanding these factors is crucial for developing effective strategies to promote deep reading in digital contexts.

This study explores the impact of digital media consumption on tertiary learners' deep reading engagement. By examining students' reading habits, preferences, and comprehension levels, the research seeks to identify how much digital reading influences their ability to engage deeply with texts. The study also intends to uncover the underlying factors contributing to observed differences in reading engagement between digital and print media.

The research will comprehensively understand the relationship between digital media consumption and deep reading through a mixed-methods approach, combining quantitative reading comprehension assessments with qualitative insights from student interviews. The findings will inform educators and policymakers how to support students in developing effective reading strategies in an increasingly digital learning environment.

The transition to digital reading presents both opportunities and challenges for tertiary learners. While digital texts offer accessibility and interactivity, they may also hinder the development of deep reading skills essential for academic success. Addressing this issue requires a nuanced understanding of how digital media consumption affects reading engagement. This study seeks to fill the existing research gap by investigating the impact of digital reading on deep comprehension among university students, ultimately contributing to developing pedagogical practices that foster critical and reflective reading in the digital age.

Research Questions:

1. How do tertiary students describe their reading engagement when interacting with academic texts compared to digital media content?

2. What factors do students perceive to influence their ability to concentrate and comprehend during extended reading tasks in a digitalised environment?

Literature Review

Digital media has revolutionised the reading habits of tertiary students, prompting a shift from traditional print to digital formats. This transition raises critical questions about how digital media consumption influences deep reading engagement, particularly in academic contexts.

Reading Engagement: Academic Texts vs. Digital Media Content

Tertiary students' engagement with academic texts often differs from their interaction with digital media content. Studies indicate that students perceive print-based academic reading as more conducive to deep engagement, allowing for better concentration and comprehension. In contrast, digital media content, characterised by hyperlinks and multimedia elements, tends to encourage skimming and superficial reading. Delgado et al. (2018) found that students reading digital texts exhibited lower comprehension levels than those reading print, particularly in tasks requiring deep understanding. Similarly, Mangen et al. (2019) reported that digital reading often reduces recall and inferential reasoning.

The cognitive processes involved in digital reading differ from those in print reading. Digital texts often encourage skimming and scanning, which can impede the development of deep reading skills. Wolf (2018) argues that the brain's circuitry for deep reading is not naturally activated when engaging with digital texts, potentially affecting critical thinking and empathy. Moreover, the constant distractions inherent in digital environments, such as notifications and hyperlinks, can fragment attention and hinder sustained focus (Carr, 2010).

Factors Influencing Concentration and Comprehension in Digital Environments

Several factors influence students' concentration and comprehension during extended reading tasks in digital environments. One significant factor is cognitive load. The presence of hyperlinks and multimedia elements in digital texts can increase extraneous cognitive load, diverting attention from the main content and impairing comprehension (DeStefano & LeFevre, 2007). Additionally, the physical aspects of digital reading, such as screen glare and scrolling, can cause eye strain and reduce reading efficiency (Margolin et al., 2013).

Another critical factor is digital self-efficacy. Students with higher confidence in their digital reading abilities are likelier to engage deeply with digital texts. Conversely, those with lower digital self-efficacy may experience anxiety and disengagement, negatively impacting comprehension (Bandura, 1997). Furthermore, the design of digital reading platforms plays a role in reading engagement. Features such as adjustable font sizes, annotation tools, and distraction-free modes can enhance concentration and comprehension (Liu, 2005).

Environmental factors also affect digital reading engagement. Students often read digital texts in environments filled with distractions, such as social media notifications and multitasking opportunities, which can disrupt focus and hinder deep reading (Junco & Cotten, 2012). Creating structured reading environments and promoting mindfulness can mitigate these distractions and improve reading engagement.

The shift from print to digital reading among tertiary students presents opportunities and challenges for deep reading engagement. While digital media offers accessibility and convenience, it also introduces factors that can impede concentration and comprehension. Understanding how students perceive reading engagement across different media and identifying the factors influencing their ability to concentrate and comprehend in digital environments is crucial for developing effective reading strategies and educational interventions. Future research should explore these dynamics to support students in navigating the digital reading landscape effectively.

Methodology

This study adopts a qualitative research design to explore how digital media consumption influences deep reading engagement among tertiary learners. A qualitative approach is most suitable for this investigation as it allows for an in-depth understanding of participants' subjective experiences, perceptions, and interpretations (Creswell & Poth, 2018). It enables the researcher to capture the richness and complexity of students' reading behaviours and attitudes in their own words, particularly in academic versus digital media reading.

The study utilises a qualitative descriptive design, which is appropriate for obtaining detailed insights into students' reading engagement without manipulating the study environment (Sandelowski, 2000). This design supports the collection of narrative data that reflects how learners make sense of their reading practices in a digitalised educational context.

The participants are ten diploma-level students enrolled at Politeknik Kota Kinabalu, Sabah. These students were selected through random sampling to ensure that each student had an equal chance of being included in the study, thereby enhancing the credibility and representativeness of the sample (Etikan, Musa, & Alkassim, 2016). All participants are currently engaged in coursework that requires print and digital reading, making them suitable informants for exploring the research questions.

Data were collected using open-ended survey questions designed to elicit comprehensive, thoughtful responses from participants regarding their reading engagement and comprehension in digital and academic contexts. Open-ended questions are particularly valuable in qualitative research, allowing respondents to express their experiences, reasoning, and attitudes in detail (Patton, 2015). These questions allowed participants to describe their preferences and behaviours and the contextual factors influencing their ability to focus and comprehend during extended reading tasks.

The survey was distributed through Google Forms, a practical and accessible tool for reaching students regardless of time and location. Google Forms facilitates anonymous participation and efficient data organisation, which is beneficial for thematic analysis. Additionally, the platform supports a user-friendly interface that encourages participation and honest responses (Wright, 2005).

The collected data were analysed using thematic analysis. Thematic analysis is suitable for identifying, analysing, and reporting patterns (themes) within qualitative data. It allows the researcher to construct a rich, detailed account of the participants' reading experiences and the factors that shape them. All responses were read multiple times to gain an overall understanding before coding began. Codes were then grouped into broader themes that reflected participants' shared experiences and perceptions. Throughout the process, the researcher ensured that themes were grounded in the data and accurately represented participants' voices.

Findings

How do tertiary students describe their reading engagement when interacting with academic texts compared to digital media content?

1. Digital Media Is More Engaging and Entertaining but Less Deep

Many students described digital media content such as blogs, social media, and online articles as more **entertaining, convenient, and easier to access**, leading to higher levels of *surface engagement*. However, this engagement was often **short-lived** and not conducive to deeper learning or retention.

"To be honest, digital content got my attention more than academic materials since it is more convenient to find information." – *Participant 7*

"I find it easier to focus on digital content because it is lighter and more engaging." – *Participant 8*

"Digital content like social media or blogs is more engaging and easier to digest, so I tend to spend more time on them without even realising it." – *Participant 4*

2. Academic Reading Requires More Cognitive Effort but Leads to Better Retention

Academic texts were consistently described as more **demanding, slower to process, and less stimulating** than digital media. However, students acknowledged that academic materials often yield **greater depth of understanding** and **longer-lasting retention**, especially when effort and active strategies (like note-taking) are applied.

"I tend to retain information from academic texts better when I take notes or highlight as I read, but it requires more effort." – *Participant 12*

"I focus better and retain information more effectively from academic texts, while digital media is more accessible but easily forgotten." – *Participant 13*

"Academic texts help me retain deeper, more detailed information, but take more time to process." – *Participant 14*

3. Reading Engagement Depends on Interest and Content Relevance

Some students emphasised that their academic or digital media engagement depends largely on the topic's personal relevance or interest level. If a topic is perceived as meaningful or relatable, they tend to engage more deeply, regardless of format.

"It depends on the academic materials. If I find it interesting, then I will focus on it more. And the same goes for digital content." – *Participant 3*

"I would say it depends… if I am in a hectic week or exam week, I find myself enjoying reading academic materials compared to digital content." – *Participant 5*

"Recently, I read an article about a civil engineering case study. I felt engaged because it was directly related to what I am studying." – *Participant 18*

What factors do students perceive to influence their ability to concentrate and comprehend during extended reading tasks in a digitalised environment?

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1. Quiet and Controlled Environments Support Concentration

Participants frequently emphasised the need for **quiet, peaceful settings** to maintain focus during long digital reading sessions. These environments often include personal spaces like bedrooms, the absence of noisy distractions, and sometimes specific sensory preferences like natural air or calm ambience.

"It would be easier for me to concentrate when I am in my room alone with classical music…" – *Participant 1*

"I will be more focused in a quiet place rather than in a crowded place." – *Participant 3*

"For me, it will get easier if the surroundings are quiet with natural air when I have time." – *Participant 10*

2. Distractions and Noise Impede Digital Reading Engagement

Students noted that **external disruptions** such as roommates, ambient noise, or device notifications often hinder their ability to focus during digital reading. These distractions, both environmental and digital, were described as barriers to sustained attention.

"There is one environment where it makes it difficult for me to concentrate during long reading, which is when the roommate… makes a loud voice…" – *Participant 2*

"Distractions like phone notifications or background noise make it much harder to stay focused, especially when reading on a screen." – *Participant 4*

"It gets difficult when there is noise, tiredness, or constant notifications from my digital devices." – *Participant 8*

3. Personal Well-being Affects Cognitive Focus

Concentration is not solely dependent on the surroundings' physical and emotional states, such as fatigue, stress, or general health, which also affect reading focus. These internal conditions can diminish the cognitive resources needed for comprehension.

"Personal factors such as fatigue, emotional distractions, or health issues can make it difficult to concentrate…" – *Participant 6*

"Being interested in the topic, having a quiet environment, and taking short breaks help me concentrate better." – *Participant 8*

"For me, I like to dive into my thoughts… it does not matter if I was alone or surrounded by people as long as it is peaceful." – *Participant 7*

4. Self-Regulation Strategies Enhance Digital Reading Focus

Participants shared **strategies and habits** for improving focus and comprehension during digital reading. These included listening to music (especially lofi or classical), using productivity methods like the Pomodoro Technique, turning off notifications, and making summaries or highlights.

“I usually use the Pomodoro method to stay focused… 50 minutes study for each session and take 10 minutes rest in between.” – *Participant 13*

“I try to take short breaks using the Pomodoro technique and make summaries or bullet points while reading. I also turn off notifications…” – *Participant 14*

"I silence notifications, use a timer, highlight key points, summarise after reading, and adjust screen settings to stay focused." – *Participant 17*

Discussion

This study explored how tertiary students engage with academic texts versus digital media content and what factors influence their concentration and comprehension in a digitised reading environment. The findings reveal a complex interplay between media format, cognitive effort, environmental conditions, and personal motivation, contributing valuable insights to the ongoing discourse on reading engagement in the digital age, particularly within English language learning contexts.

The results show that students found digital media content more accessible and entertaining, and more superficial regarding cognitive engagement. Participants frequently described digital reading as light and quick, with little retention or deep understanding. This aligns with prior research indicating that digital texts, while convenient, often foster surface-level processing due to multitasking and hyperlinked environments (Delgado et al., 2018; Mangen et al., 2019). The cognitive habits formed in digital contexts, such as skimming and browsing, contrast the deep reading processes required for academic success, as Wolf (2018) highlighted.

In contrast, academic texts were perceived as more cognitively demanding but ultimately more beneficial for comprehension and knowledge retention. Students acknowledged that such reading required greater effort, note-taking, highlighting, and re-reading, but resulted in more meaningful engagement. This supports Guthrie and Wigfield’s (1997) assertion that intrinsic motivation and strategic learning behaviours underpin deeper literacy practices. It also reflects the findings of Clinton (2019), who argued that overconfidence in digital reading often masks comprehension deficits.

Interestingly, several participants noted that reading engagement depended heavily on personal interest and content relevance, regardless of whether the material was in digital or print format. This suggests that autonomy and meaningful content are critical mediators of reading motivation, an idea echoed by self-determination theory (Briggs, 2009) and research by Konrad (2023), who found that student choice and identity alignment in reading materials significantly enhance motivation.

Another important set of findings concerned the factors affecting concentration and comprehension in digital environments. Students consistently identified quiet, controlled environments as crucial for sustained focus. Distractions such as ambient noise, device notifications, and social interruptions were cited as detrimental. These observations are consistent with Carr (2010) and Junco and Cotten (2012), who found that digital multitasking environments significantly reduce attention span and cognitive endurance.

In addition to environmental concerns, personal well-being, including stress, fatigue, and emotional state, emerged as a key influence on reading engagement. This echoes Wolf’s (2018) suggestion that digital reading is not only a cognitive process but also an embodied and emotional one. Furthermore, students discussed self-regulation strategies such as the Pomodoro Technique, screen adjustments, and digital annotation. These practices reflect a growing metacognitive awareness among learners about optimising their focus and comprehension in digital settings.

This study contributes new insights by illuminating how students negotiate the trade-offs between convenience and depth in digital reading environments. At the same time, much literature compares outcomes between digital and print reading and fewer studies centre on students' voices in exploring the nuances of engagement. This research reveals a more nuanced reality: students are not passive digital consumers. However, they are aware of the limits and affordances of digital reading and are developing personal coping strategies.

Moreover, by highlighting students' self-awareness and adaptive reading behaviours, this study challenges deterministic narratives that digital reading always leads to cognitive decline. Instead, it presents a more agentic view of learners who can engage deeply with texts even in digital environments when equipped with strategies and supportive conditions.

Another unique contribution lies in its context. The study within a Malaysian polytechnic adds much-needed diversity to the largely Western-centric discourse on digital reading, offering insights shaped by regional, technological, and educational conditions.

Implications of the Study

These findings have significant implications for English language learning (ELL) educators. Firstly, academic reading in English requires linguistic proficiency, cognitive stamina, and focused environments. Educators must explicitly teach deep reading and comprehension strategies as students are increasingly expected to consume English texts digitally, especially in blended and online learning.

Secondly, the study underscores the importance of relevance and student interest in selecting English reading materials. ELL instructors should prioritise culturally responsive texts and offer choices when possible, aligning with evidence that student engagement rises when texts resonate with learners' lived experiences (Gilson et al., 2018).

Finally, it is vital to incorporate metacognitive and self-regulation training into English reading curricula. Teaching students how to manage distractions, annotate texts, and reflect on comprehension can significantly improve reading outcomes. Such pedagogical interventions are especially necessary in multilingual classrooms, where students face cognitive load from language decoding and digital navigation.

Limitations of the Study

While this study provides rich insights, several limitations must be acknowledged. First, the sample size was limited to ten students from a single institution, which may restrict the generalisability of the findings. Second, the study relied on self-reported data, which may be influenced by social desirability or memory biases. Third, while the study focused on students' perceptions, it did not directly measure comprehension outcomes, which could further substantiate claims about deep reading engagement.

The study did not differentiate between digital formats (e.g., PDF textbooks, e-books, or social media posts), which may offer different affordances and constraints. Future studies could use mixed methods or experimental designs to triangulate findings and assess the impact of specific digital media types on comprehension and retention.

Contribution to the Field

This study contributes to the broader discourse on digital literacy by demonstrating how tertiary students critically engage with their reading environments and adopt strategies to sustain deep reading amidst digital distractions. The field of English language learning offers a timely reminder that literacy in the 21st century must include both technical digital fluency and cognitive reading resilience.

As digital reading is increasingly embedded in higher education, ELL educators must move beyond binary debates about print versus screen. Instead, the focus should shift toward empowering students with skills, awareness, and environments that support deep and reflective reading, regardless of medium. In doing so, educators can better prepare students for academic success and lifelong engagement with the language and meaning that underpins English literacy.

Conclusion

This study examined how tertiary students engage with academic texts compared to digital media content and explored the factors influencing their concentration and comprehension in a digitalised reading environment. The findings highlight a nuanced understanding of reading engagement among students, revealing that while digital media is perceived as more accessible and entertaining, it often lacks the cognitive depth required for academic learning. Conversely, academic texts, though more demanding, are associated with better retention and deeper understanding, particularly when supported by active reading strategies and a conducive environment.

Students' reading engagement is shaped not only by the medium of the text but also by personal interest, emotional well-being, environmental factors, and self-regulation practices. This underscores the need for a holistic approach to supporting reading in higher education, considering the interplay between digital infrastructure, learner agency, and pedagogical practice. The study affirms the importance of fostering digital literacy and deep reading skills for educators and practitioners in English language learning. Incorporating student-centred strategies, promoting metacognitive awareness, and designing distraction-free digital learning environments are essential to enhance reading engagement. Additionally, empowering students to select meaningful and relevant texts can help sustain motivation and deepen comprehension, especially in multilingual and multicultural classrooms.

While this study is limited by its small sample size and qualitative scope, it provides valuable insights into students' lived experiences navigating the challenges of digital reading. Future research should build on these findings by exploring digital reading engagement across diverse contexts and integrating direct comprehension and cognitive effort measures. Ultimately, promoting deep reading in the digital age is not choosing between print and screen but equipping learners with the tools, environments, and mindsets necessary to read with focus, purpose, and understanding.

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